~~Logistic Regression~~

~~Gradient Descent~~

- Logistic Regression on m examples

$J = 0 \; ; \; dw_1 = 0 \; ; \; dw_2 = 0 \; ; \; db = 0$

For $i = 1$ to m

$$z^{(i)} = w^T x^{(i)} + b$$

$$a^{(i)} = \sigma(z^{(i)})$$

$$J \mathrel{+}= -[y^{(i)} \log a^{(i)} + (1-y^{(i)}) \log(1-a^{(i)})]$$

$$dz^{(i)} = a^{(i)} - y^{(i)}$$

$$\left.\begin{aligned} dw_1 &\mathrel{+}= x_1^{(i)} dz^{(i)} \\ dw_2 &\mathrel{+}= x_2^{(i)} dz^{(i)} \end{aligned}\right\} \rightarrow \text{\# of features is 2}$$

$$db \mathrel{+}= dz^{(i)}$$

$J \mathrel{/}= m$

$dw_1 \mathrel{/}= m \; ; \; dw_2 \mathrel{/}= m \; ; \; db \mathrel{/}= m$

$w_1 := w_1 - \alpha \, dw_1$

$w_2 := w_2 - \alpha \, dw_2$

$b := b - \alpha \, db$

Here, we use two for loops, but with higher ~~total~~ quantity of data (millions), we need our program to be more efficient, so we use vectorization